Management of mental health emergencies in elite athletes: a narrative review

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ABSTRACT

Mental health emergencies require a rapid, effective response. We searched the literature on mental health emergencies in athletes and found five papers. None of these addressed elite athletes. Nonetheless, common mental health emergencies may present in the sports environment and may place the athlete and others at risk. Sports teams and organisations should anticipate which emergencies are likely and how medical and support staff can best respond. Responses should be based on general non-sporting guidelines. We stress the importance of clinicians following standard procedures.

INTRODUCTION

A mental health emergency exists in elite athletes when the individual presents with an acute disturbance in mental state associated with either an underlying mental health or other medical disorder. 'Acute disturbance' includes features such as agitation, aggression and violence, although these terms are not defined or operationalised in the context of describing an emergency.[1 2] Features of an emergency presentation also include a significant level of immediacy and risk.[2 3] The risk may be to the athlete or to others in the vicinity of the athlete; importantly, the individual suffering with a mental health emergency often has impaired insight. Risks include those arising from suicidal or homicidal thoughts and behaviours, or from impaired judgement and recklessness such as in mania.[3]

Clinicians who work with athletes need guidance on an appropriate response, especially as emergencies may arise when an athlete does not have rapid access to experienced mental health personnel or appropriate facilities. Evidence-based guidance for elite athletes is based primarily on extrapolation from more general (and non-sporting) guidelines and 'expert opinion'. In this narrative review, we synthesised the available literature on mental health emergencies in athletes and complemented it with evidence-based and consensus-based recommendations.

METHODS

Key databases (PubMed, SPORTDiscus, PsycINFO, Scopus and Cochrane) were searched for English-language papers by one author (MEH) at the start of the project and 3 months before intended submission. First, search terms relating to mental health emergencies were combined with terms relating to sport participation. Using this strategy, 96 articles were identified; after screening for relevance, this was reduced to 5. When these terms were combined with terms reflecting elite athlete status, no results were found. Elite athletes were defined as those competing at professional, Olympic or collegiate/university levels. The five core references were then used by all authors to identify supplementary material. Finally, the non-sporting literature was reviewed for relevant papers on diagnosis, prevalence and treatment in mental health emergencies.

MENTAL HEALTH EMERGENCY PRESENTATIONS

Mental health emergencies may present in elite athletes through the entire range of mental health symptoms and disorders, including neurocognitive disorders such as delirium, substance use, bipolar and psychotic disorders, depression, suicidality and self-harm, anxiety, eating disorders, and personality disorders.

At the London 2012 Olympics, a dedicated psychiatry and psychology service was provided to the more than 10 000 Olympic and Paralympic athletes by Cognacity Wellbeing via the Olympic Polyclinic. Of the 12 cases referred (11 athletes, 1 administrator), 4 constituted a mental health emergency, defined as an acutely disturbed mental state resulting from a mental health disorder and associated with a high level of risk. Among these, two appeared to meet the criteria for adjustment disorder, one for panic disorder and one for post-traumatic stress disorder (unpublished IOC/Cognacity data).

Delirium

Delirium is a neurocognitive disorder and the direct physiological consequence of another medical condition, substance intoxication or substance withdrawal. Delirium presents as an abrupt change from normal baseline mental functioning[4]

In elite athletes case reports describing delirium after head injuries have been documented.[5] Both hyponatraemia and hyperthermia may present as delirium.[6] Although exercise-associated hyponatraemia was initially described primarily in endurance events such as ultramarathons,[7] it is now recognised in a wide range of sports. Overzealous hydration is a key factor.[8]

Substance misuse

Athletes may use or misuse illicit and legal drugs to enhance performance or cope with stressors.[9 10] Such behaviours may contribute to a higher risk of mental health emergencies via intoxication, delirium or withdrawal, and these states may include adverse reactions such as psychosis. Substance use disorders and associated problems (including withdrawal

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syndromes) are not restricted to active elite athletes and may be more common in retired athletes.[11–13]

Binge alcohol drinking may be more common among athletes than non-athlete peers.[14–16] Thus, athletes may be more likely to present with acute intoxication and its complications rather than dependence/withdrawal. Nonetheless, heavy regular drinking has been reported in elite professional athletes[17–19] and US collegiate athletes.[20 21] Alcohol is an important moderator of the suicidal ideation and violent behaviour sometimes associated with use of anabolic androgenic steroids (AAS).[22 23]

Cannabis use is common among elite athletes in some sports, for example, snowboarding and lacrosse.[24 25] Delirium and aggression have been reported in this context, and misuse of high-potency cannabis may activate acute anxiety or even trigger psychotic symptoms.[26 27] This situation is especially likely if the cannabis contains high levels of tetrahydrocannabinol or is contaminated with synthetic cannabinoids.[26] However, when use by elite athletes is 'recreational',[24] anxiety and dysphoria effects are usually mild, although paranoid symptoms can emerge.[28]

Androgenic anabolic steroid use is linked to hypomanic or psychotic symptoms, aggression, and even suicide.[22 29 30] Symptoms can emerge within days,[31] and higher doses and use of multiple agents are associated with greater mental health disturbances.[22 29 32] There may be two patterns of suicidality related to AAS use[33]: current users may experience rapid mood changes associated with impulsive acts, while former users can develop prolonged mood changes and depressive disorders during withdrawal associated with more sustained suicidal thinking, intent and planning.[33] Both scenarios carry significant risks, and death by suicide is two to four times greater in male athletes who have used AAS compared with the general population.[34]

Stimulants such as amphetamine, d-methamphetamine, ephedrine, pseudoephedrine, caffeine, nicotine, methylphenidate, modafinil, cocaine and strychnine may be used or misused by athletes.[9 35] Use of these stimulants may lead to insomnia, agitation, aggression and psychosis, especially if individuals are 'stacking', meaning using several stimulants concurrently.[9 10 36]

In the UK, opioids are the most common drugs used in self-poisoning and account for 33% of self-poisoning fatalities (https://sites.manchester.ac.uk/ncish/). Opioid intoxication is a concern in sport because of the use of non-prescribed opioids in athletes. Eighteen per cent of injured male US collegiate athletes reported non-medical use of prescription opioids in the past 12 months compared with 8.3% of college students in general.[37] In addition, 11% of US collegiate athletes reported using prescribed opioids in the previous 12 months, along with 3% using similar medication without a prescription.[25]

Bipolar and psychotic disorders

There are no available data on the prevalence of bipolar and psychotic disorders in elite athletes[38–40]; these disorders can present as a mental health emergency. Examples include impaired judgement in mania, impacting on activities such as spending, driving or sexual activity,[41] or severe agitation and disorganised behaviour resulting from psychotic symptoms such as delusions or hallucinations.[42]

Depressive disorders

Depressive disorders in elite athletes may be as common as in the general population, or in some circumstances even more common.[17 38–40 43–46] Depressive disorders may promote suicidal thinking, intent and behaviour.

Sports injuries are common[47 48] and are associated with depressive symptoms and disorders.[40 44 48–50] The relationship between sports injury and mood is not simply cause and effect. Stress increases injury risk, while the emotional reaction to injury affects not only mood but thinking and behaviour and can prolong recovery.[51] Furthermore, sports injuries are associated with psychological variables such as anger, which can increase suicide risk.[52 53]

Depressive symptoms may go unacknowledged in elite athletes because of an assumption that elite athletes are mentally 'tough'.[54] Further, depressive symptoms can be erroneously interpreted as overtraining rather than a depressive disorder.[55] This may increase the possibility of a mental health emergency presentation, with symptoms unrecognised until a crisis emerges.

Enforced retirement, or time away from sport for reasons of injury and illness, also increases distress levels[12] and the risk of emotional decompensation.[56] Deselection (being dropped from a team) is associated with adverse emotional reactions; in a prospective study, over half of players whose contracts were not renewed reported clinical levels of psychological distress, including symptoms of depression, anxiety, loss of confidence and social dysfunction.[57]

Suicidality and self-harm

Suicide is the second most common cause of premature mortality in adults ages 15–29 in the general population,[58] coinciding with the peak ages of elite performance for many athletes. Suicide rates are approximately three times greater in the general population of men compared with women,[59] and systematic under-reporting has been noted.[60] The most commonly used methods of death by suicide within the general population are hanging, self-poisoning, ingestion of pesticides and firearms, but reporting among different countries is highly variable.[59]

Suicide rates are lower among male college students in the USA who play sports and are physically active.[61] Conversely, college students who do not participate in sports have a higher risk of suicidal behaviour.[62] However, multiple risk factors such as injury, performance pressures and substance misuse may be present in athletes,[63] who may also be more susceptible to risk-taking behaviours in general.[40] Suicide risk seems greater after retirement from sport compared with during active sport participation.[34 64]

Non-suicidal self-injury is characterised by intentional self-inflicted damage to the body, often in response to negative thoughts and feelings.[4] Self-harm is common in young people, with reported rates of 7%–14% in those aged 15–19 years old.[65] Ten per cent of children participating in sport report at least one act of self-harm.[66] The phenomenon is distinct from, but related to, suicide. The National Confidential Enquiry in the UK (https://sites.manchester.ac.uk/ncish/) has identified an increased relative risk of suicide in those who have self-harmed; in 29% of all deaths by suicide, self-harm has occurred in the preceding 3 months.

Anxiety and related disorders

Unexpected, abrupt and severe acute anxiety as experienced in panic attacks or full panic disorder may be the most likely anxiety disorder to present as a mental health emergency. Available self-report data suggest that panic disorder may occur in approximately 4.5% of elite athletes, with around 1 in 10 reporting a full or limited symptom panic attack in the previous 7 days.[44] Other types of anxiety that may present acutely and require urgent attention are intense performance anxiety or

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Box 1 Features of Borderline Personality Disorder[4]

Borderline personality disorder (five or more of nine features).
1. Frantic efforts to avoid real or imagined abandonment.
2. Unstable and intense relationships. Alternate between extremes of idealisation and devaluation.
3. Markedly unstable self-image or sense of self.
4. Impulsivity in at least two damaging areas (eg, spending, sex, substance use, reckless driving, binge eating).
5. Recurrent suicidal or self-mutilating behaviour or threats.
6. Marked reactivity of mood usually lasting hours or occasionally a few days (eg, dysphoria, irritability, anxiety).
7. Chronic feelings of emptiness.
8. Inappropriate, intense anger or difficulty controlling anger.
9. Transient, stress-related paranoid ideation or severe dissociative symptoms.

flying phobias in elite athletes obliged to travel often and who may have had a fear-inducing flying experience.[67] As noted above, acute or chronic use of or intoxication with cannabis is also seen in elite athletes and may trigger acute anxiety.[24 27]

Eating disorders

Eating disorders in elite athletes are more common than in the general population, especially in weight category, aesthetic, antigravity and endurance sports.[68 69] Immediate risks among those with eating disorders usually occur as the result of other medical issues,[70] for example, electrolyte abnormalities such as hypokalaemia, refeeding syndrome, cardiac arrhythmias and skeletal fractures. In these circumstances, it can be necessary to exclude the athlete from some or all physical activity (including important competitions) until the risk is further assessed or begins to decline.[70 71] Such exclusion carries a risk of emotional decompensation and may necessitate an urgent mental health evaluation.[56]

Personality disorders

A personality disorder is diagnosed when there is an enduring and pervasive pattern of inner experience and behaviour that is inflexible, maladaptive, and leads to significant distress or functional impairment. Limited data point to borderline and obsessive-compulsive personality disorders as possibly the most common personality disorders among competitive athletes.[72]

The features of borderline personality disorder are listed in box 1, and several of these increase the likelihood of a mental health emergency presentation. Those with borderline personality disorder often experience unstable and intense relationships that alternate between idealising and then devaluing other people.[4]

An impending relationship separation (including from a team, coach, clinician, other members of the athlete support staff or others) can produce profound changes in affect and behaviour, and efforts to avoid actual or perceived abandonment might include inappropriate anger and impulsive actions such as self-mutilation or even suicidal behaviour.[4] This may have particular relevance in a sporting context if the athlete is dropped from a team/fails to qualify or has a contract or funding terminated.

Since borderline personality disorder traits are often comorbid with eating disorders,[73] exclusion and deselection issues that arise in this context can be problematic. Individuals with borderline personality disorder are also frequently impulsive and engage in risk-taking behaviours[4] that can result in an emergency presentation (see box 1 for examples of areas of impulsivity). An unstable affect is an additional common feature of borderline personality that manifests as an extreme reactivity of mood often in response to interpersonal stressors.

Borderline personality disorder leads to high levels of use of mental health and other healthcare services, and the impairments and suicide risk associated with the condition are highest in young adults in their 20s and begin to decline at age 30.[4]

Personality traits such as the intensely emotional and erratic ones found in borderline personality disorder are seen in many AAS users and may contribute to the excessive violence, aggression and mood changes seen in some AAS users.[22]

Table 1 Mental health assessment in an emergency

Presenting problem.	A multisource account of the current crisis and its symptoms and/or behaviours of concern.
History of problem.	Current and past stressors. ► Including recent losses or traumatic experiences.
Recent substance use.	Recent substance use history.
Support.	Support systems. ► Including family, team mates and key members of support staff.
Strengths/weaknesses.	Personality style including strengths and weaknesses. ► Includes previous helpful and unhelpful coping strategies.
History.	Current/past medical and psychiatric history.
Mental state examination.	Detailed mental status examination. Observe and document appearance, behaviour, speech and thought content and form, affect/mood, any perceptual abnormalities, cognitive impairments (especially awareness and orientation), and the athlete's insight and understanding of the problem.
Risk.	Assessment of risk of self-harm, aggression and other risks related to impaired judgement.
Context.	Assessment of the training, team, family and cultural context in which the emergency arose. May be especially important during recovery and rehabilitation and in a postemergency debrief.
Rating scales.	Consider routine use of rating scales. For example: ► Anxiety: GAD-7.[104] ► Depression: PHQ-9.[105] ► Suicidality: SBQ-R. ► Aggression: Symptom Screening Form.
Investigations.	Directed by examination findings. Especially important if delirium is suspected. Should include a toxicology screen for common substances of misuse (such as stimulants, sedatives, hypnotics, muscle relaxants, opiates, alcohol, cocaine, cannabis and synthetic cannabinoids, anabolic steroids, fentanyl, methylenedioxymethamphetamine, lysergic acid, and phencyclidine).[67 74 75 104–106]

GAD, generalised anxiety disorder; PHQ, Patient Health Questionnaire; SBQ-R, Suicidal Behaviors Questionnaire-Revised.

ASSESSING THE ATHLETE FOR A MENTAL HEALTH EMERGENCY

Assessment of elite athletes experiencing a mental health emergency should follow a structured approach as in any other mental health assessment.[74–76] This assessment begins with a thorough history, ideally including collateral input from team mates, athletics staff and stakeholders, and family. Key components of the assessment are listed in table 1.

The values that guide an appropriate emergency response are also important. These include (1) avoiding physical and

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emotional harm; (2) using person-centred approaches with shared responsibility, respect and use of an individual's strengths and natural support networks; (3) establishing safety; and (4) adhering to principles of hope, recovery and resilience.[77]

Service provision and principles of care

The foundation of providing service for a mental health emergency is to have in place a written, rehearsed mental health emergency action plan (http://www.ncaa.org/sport-science-institute/mental-health-best-practices). Such a plan must account for mental health emergencies in all situations, including travel. In addition, provision of emergency mental healthcare requires both suitable facilities and clinical expertise. A suitable facility or environment in which to provide care in an emergency should balance safety with containment, be uncluttered and provide emergency escape routes.[3 78] Such facilities can be harder to access when providing medical care away from the clinic at a training or competition venue.[74] Even with proper planning, mental health emergencies often arise unexpectedly, and evaluations may need to occur in the team hotel, training room or athlete's residence. In these situations, it is often prudent to have other staff such as a sports physiotherapist or team physician sit in or be immediately available for support and consultation.[79] It is important to be familiar with local facilities either close to the team base or when travelling. This includes familiarity with local arrangements and legislation for statutory detention.[80]

For athletes with mental health symptoms or disorders, there should be an identified clinician with appropriate training and expertise to provide ongoing treatment. This may begin with a sports physician allied to the team or sport, and then should include a psychiatrist and/or clinical psychologist as necessary.

Access to mental health expertise is likely to be required quickly in a mental health emergency. In the USA, the National Collegiate Athletic Association best practice guidelines (http://www.ncaa.org/sport-science-institute/mental-health-best-practices) emphasise the importance of defining what constitutes an emergency and the need for clarity on who is competent to intervene, how they should be contacted and when (box 2). The guidelines also emphasise the importance of anticipation—preplanning how to respond to an emergency, and reviewing previous crisis events to improve services and action plans. Plans should be consistent with other emergencies in sport, for example, cardiac arrest, heat illness and severe physical trauma.

Box 2. Core elements of a mental health emergency action management plan

Identify situations, symptoms or behaviours that are considered mental health emergencies.

Develop written procedures for management of the following mental health emergencies:

- Suicidal or homicidal ideation.
- Sexual assault.
- Highly agitated or threatening behaviour, acute psychosis or paranoia.
- Acute delirium/confusional state.
- Acute intoxication or drug overdose.

Specify situations in which the individual responding to the crisis situation should immediately contact emergency medical services.

Be certain that individuals responding to the acute crisis are familiar with local municipality protocols for involuntary retention, for example, if the athlete is at risk of self-harm or harm to others.

Specify situations in which the individual responding to the crisis situation should contact a trained on-call counsellor.

Adapted from http://www.ncaa.org/sport-science-institute/mental-health-best-practices.

MANAGEMENT

Delirium

Guidance on acute treatments for mental health emergencies emphasises a hierarchy of interventions.[1 2 81] As almost any medical condition can lead to delirium, the primary issue is to identify any underlying conditions and to treat appropriately.[81] Sports-related concussion typically results in only short-lived impairments and is managed by a variable period of cognitive and physical rest.[82] The standard treatment for exercise-associated hyponatraemia is intravenous hypertonic saline, although occasionally oral administration may be tried.[8] Cold water immersion is used to treat exertional heat stroke.[83]

Substance use: intoxication and withdrawal

The most common intoxicants that may result in a mental health emergency in elite athletes are alcohol, cannabis/synthetic cannabinoids, stimulants and opioids.[14 25 84 85] Alcohol intoxication is most likely to present to clinical attention following a blackout, arrest or fight/injury. For safety, it is important to monitor the athlete in a controlled setting, such as a training facility, team mate's or family member's residence, or emergency room. This will help to prevent further dangerous behaviours, such as additional drinking, driving, aggressive/self-injurious behaviours, cold exposure or aspiration of vomit.[86] A comprehensive assessment within a day or two (to allow time for intoxication to clear) is then necessary. This includes assessing details of the incident and the recent drinking pattern. The next step is developing a treatment plan that matches the severity of the problem. The plan could include counselling, psychotherapy, more intensive treatment with team or family involvement, and a recommended period of abstinence.[87 88]

Cannabis/synthetic cannabinoid intoxication is most likely to present with panic, disorientation, perceptual distortions (illusions or hallucinations) or paranoia.[26] In some cases, the disturbance is mild and self-limiting and may be managed with support and observation, for example, at the team hotel. Other cases will require a more detailed clinical assessment in an emergency room, and may require hospitalisation with comprehensive mental health and other medical assessment.

Stimulant intoxication is most likely to result from chronic, high-level use of one or more stimulants, including medication used for attention-deficit/hyperactivity disorder (taken with or without a prescription), ephedrine or other 'exercise enhancers', nicotine, caffeine, or cocaine.[10 25 86] Presentations can include heat illness, anxiety, agitation, irritability, anorexia and sleeplessness.[10] Treatment begins with emergency management of heat illness (when present), then identification of the number and dosages of the stimulant(s), and a prompt reduction of dose or elimination of one or more drugs. Short-term, low-dose benzodiazepines or hypnotics may be necessary and should be integrated as necessary with motivational therapy for behavioural change.[10 86 89 90]

Non-prescribed opioid use is common in sport.[37] In addition when athletes ingest opioids that are not manufactured in a pharmacy, there are concerns about crossover to heroin or experimentation with cocaine that might be contaminated with

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fentanyl or other potent synthetic opioids.[37] Acute opioid intoxication with lowered level of consciousness and reduced respiration is a medical emergency that can be treated with narcotic antagonists such as naloxone.[91] Some opioids have much longer durations of actions than others, so repeated administration of the antagonist may be necessary. As with other substance-related emergencies, a comprehensive assessment and treatment plan should be developed within a few days.

Androgenic anabolic steroid use may lead to brief and self-limiting mental state disturbances,[32] and treatment may only be needed for days or weeks, unlike cases of bipolar disorder and psychosis. Benzodiazepines, antipsychotics and occasionally valproate salts have been suggested for use.[92]

Bipolar disorder

Mania may occasionally present as a mental health emergency when there are risks associated with impaired judgement in tandem with an elevated mood. These may be so severe that inpatient treatment is necessary.[93] If judgement is impaired to the point where capacity is lacking, statutory detention may also be necessary,[93] thus highlighting the importance of awareness of local arrangements for this possibility when travelling with a sports team.[80] Acute management of the associated behavioural disturbance follows the de-escalation and pharmacological strategies outlined below. Pharmacological treatment usually involves benzodiazepines or antipsychotics alone or in combination, with the former often preferred if the diagnosis is uncertain.[1]

Psychotic disorders

An acutely psychotic individual may be highly agitated by delusions or hallucinations and/or highly disorganised to the point where they are unable to care for themselves.[42] In these circumstances, inpatient care may be required and under statutory arrangements if necessary.[42] The associated behavioural disturbance may need to be managed by de-escalation techniques,[78] augmented if necessary by calming medication.[1]

Suicidality and self-harm

In a suicidal patient or after an act of self-harm, multidisciplinary care using a biopsychosocial framework reduces risk of suicide[94] and self-harm (https://www.nice.org.uk/guidance/cg133). Clinicians may worry that asking about suicide will initiate suicidal thoughts or actions, but no data support this concern.[95] Clinicians should ask directly about suicide and homicide, including thoughts, intent and/or plans and listen to the athlete's concerns.[95]

Emergency services should be contacted quickly if needed (this requires knowledge of facilities at current location) and other immediate medical healthcare should be provided as appropriate, for example, basic life support or referral for trauma care.

It is also important to ensure the immediate safety of the environment for the athlete and the clinician, for example, carefully remove bladed items and poisons/toxins from the athlete's possession.[95]

Good practice extends to supporting the athlete's ability to avoid future suicidal behaviour, including restricting access to means of harm, for example, medicines that may be used in self-poisoning.[95] It is helpful to discuss a safety plan, which may be a simple written document that specifies how the athlete can manage any suicidal thoughts, for example, by contacting supportive people who have been found helpful in the past. However, do not rely solely on a safety plan as assurance that an athlete is safe.[95]

Box 3. Acute management of suicide/self-harm risk[95 107 108]

- Directly ask about suicide and homicide (thoughts, intent and plans).
- Listen to the athlete's concerns.
- Suicidality rating scales are not needed. None are associated with a high predictive value.
- Ensure immediate safety of the environment for the athlete and the clinician.
- Contact emergency services if needed.
- Provide medical healthcare as appropriate.
- Consider whether a mental health disorder or substance intoxication/withdrawal is present.
- Assess the athlete's history of harm to self or others.
- Take steps to restrict future access to means of harm.
- Provide the athlete with hope.
- If travelling, consider support when the athlete returns home.
- Support the athlete's ability to avoid future suicidal behaviour.
- Direct the athlete towards sources of support, which may be provided by the following:
  - Healthcare staff in local medical facilities.
  - Team medical staff.
  - Team psychology/psychiatry staff.
  - Team mates, coaching and support staff.
  - Family and friends.
  - Clinicians available remotely, for example, video messaging.

To deliver such care, there should be ready access to appropriate expertise (http://www.ncaa.org/sport-science-institute/mental-health-best-practices), which is not always available in all sports venues or countries.[80] General and specific principles in the acute management of suicide and self-harm risk are illustrated in box 3.

Anxiety and panic attacks

Specific behavioural techniques to rapidly reduce arousal, such as controlled breathing or grounding techniques, may be required to treat severe acute anxiety states.[67] Occasionally pharmacological approaches are used, but benzodiazepines are likely to impair athletic performance and are more useful as short-term treatments for rapid symptom reduction of overthinking, agitation and insomnia.[67] Beta-blockers might also be helpful but may require a therapeutic use exemption[96] and may compromise both blood pressure[97] and cardiopulmonary capacity.[98]

Eating disorders

The decision to exclude an athlete from training or competition can result in acute emotional decompensation and a need for emotional support or even an urgent evaluation by a mental health clinician.[56] However, with good practice this risk can be managed and exclusion decisions need not be an abrupt surprise to the athlete. A process of encouraging the athlete to seek help is more likely to have a favourable outcome, with exclusion only as a last resort.[99] For elite athletes, the medical consensus is also to have explicit policies on disordered eating and energy deficiency that are clear on exclusion and return-to-play decisions.[70] Such policies should greatly reduce the likelihood of having to make an urgent, unexpected decision that could precipitate an emergency.

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Personality disorders

The impairments associated with borderline personality disorder tend to wane with age, but are still likely to be present in early adulthood.[4] The mainstay of treatment is longer term (more than 3 months) psychotherapy.[100] The behavioural disturbance associated with rapid mood changes that are a feature of the condition can be managed with psychological de-escalation techniques; guidance emphasises a calm, empathic approach that may be augmented with calming medication.[1 78 100]

MANAGING ACUTE BEHAVIOURAL DISTURBANCE

The management of behavioural disturbance (that may arise in any mental health emergency) begins with attempts to de-escalate the situation, augmented if necessary by oral medications, which typically should be considered before parenterally administered drugs.[1] Occasionally, physical restraint may be used but rarely, if ever, is mechanical restraint indicated.[81]

De-escalation

Verbal de-escalation can reduce the need for more invasive measures, such as medication and even physical restraint.[78] Verbal de-escalation is a collaborative approach where the focus is on actively helping patients to calm themselves. There are three steps in this process: (1) verbally engage the patient; (2) establish a collaborative relationship; and (3) verbally de-escalate.[78]

Verbal engagement begins with clinicians introducing themselves politely and clearly, providing orientation and reassurance, and speaking in concise, simple sentences.[78] Interaction with one clinician rather than several is often helpful, as multiple verbal interactions can be confusing and lead to further agitation.[78] Developing collaboration with an agitated patient involves active listening, conveyed in speech and body language, to identify the patient's needs and feelings.[78] Coaching the patient on how to regain control of their emotions can then proceed and may involve simple advice, for example, to find a quiet, comfortable space to sit and calmly relate their concerns. An offer of warmth, food or drink can also facilitate management of the situation.[78] Once the patient is calm, the clinician should provide time for debriefing. This begins with an explanation of events by the clinician to the patient, then allowance for the patient to describe events from their perspective, and concludes with an opportunity for the patient and the clinician to jointly reflect on what might be helpful in any future situations.

Pharmacological treatments for acute disturbance

In making sport-specific recommendations for pharmacological interventions, there are few data to guide treatment choice[101] other than 'careful individualised prescribing'.[102] This encompasses consideration of side effects that may negatively impact performance (eg, extrapyramidal motor side effects, sedation or weight gain).[101 102] However, these concerns may be less relevant in an emergency, where the risk of withholding treatment is higher, and only short-term treatment may be needed (days or weeks).

When medication is required, benzodiazepines are preferred if the patient has not previously had antipsychotic medication, has cardiovascular disease, if only limited clinical information is available, or the individual is not definitively known to have a bipolar or psychotic disorder. Lorazepam is most commonly recommended.[1] Other antipsychotic medications such as haloperidol, risperidone, olanzapine or droperidol can be used alongside or as an alternative to benzodiazepines,[1] especially if a patient presents with mania or psychosis.[2] Midazolam via the buccal route or inhaled loxapine can also be used.[1] Availability of specific medications depends on geographical location.

Parenteral administration may be considered if a high level of acute disturbance remains and de-escalation and oral pharmacological methods are ineffective. Use of the parenteral route (usually by intramuscular injection) is often termed 'rapid tranquilisation' or 'RT'. As with the oral route, benzodiazepines such as lorazepam or antipsychotics such as olanzapine, droperidol or aripiprazole can be used. Intramuscular promethazine (a sedating antihistamine) can also be considered. Parenteral diazepam is not recommended, as evidence is lacking. Midazolam is also not recommended because of the risk of respiratory depression.[1]

What is already known

- Mental health emergencies are situations of high risk and demand an urgent response.
- A wide range of mental health symptoms or disorders may present as an emergency.
- Recommendations for the immediate management of mental health emergencies include screening for possible causes of delirium and prompt management of acute behavioural disturbance.

What are the new findings

- There are very limited data on the nature and frequency of mental health emergency presentations in elite athletes.
- Guidelines and standard assessment and management processes for mental health emergencies in the general population can be adapted for management strategies for elite athletes.
- Better data could inform service development, improve the treatment of elite athletes in mental health emergency situations and reduce associated risks.

SUMMARY

Some sport-specific circumstances may confer an increased risk of an emergency mental health presentation. These include deselection, factors associated with sports injuries, and use and misuse of performance-enhancing drugs and other substances. There are no sports-specific treatments for mental health emergencies or for the acute behavioural disturbance that can be associated with these situations. Thus, the clinician must extrapolate evidence and recommendations from other populations.

A mental health emergency action plan is essential. Sport stakeholders (eg, organising committees of major events) should develop and rehearse such a plan that anticipates what personnel and facilities would be needed in a mental health emergency, and how these would be accessed. More accurate and systematic data collection on the range and number of mental health emergencies at major events and within sports organisations could more clearly inform such action plans. Better accessibility and quality of clinical services for mental health emergencies would allow athletes to return to health and to sport more quickly and safely.

Sports organisations should consider conducting multidisciplinary reviews after any mental health emergency to ensure valuable lessons are learnt from each case. In addition, procedures that allow staff to report events easily can enhance the value of reporting systems.[103] Sports organisations should also consider developing systems for healthcare staff to report

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centrally incidents such as mental health emergencies. Aggregated data on the nature and frequency of these events could contribute to improved services for athletes and might save lives.

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